

Topics

◦ Finish Countability

- $\mathbb{Q}$ is countable
- $\mathbb{R}$ is uncountable (decimal proof)

◦ Metric Spaces

- Definition
- Examples

Def A sequence of elements in a set A is a function $\phi: \mathbb{N} \rightarrow A$.

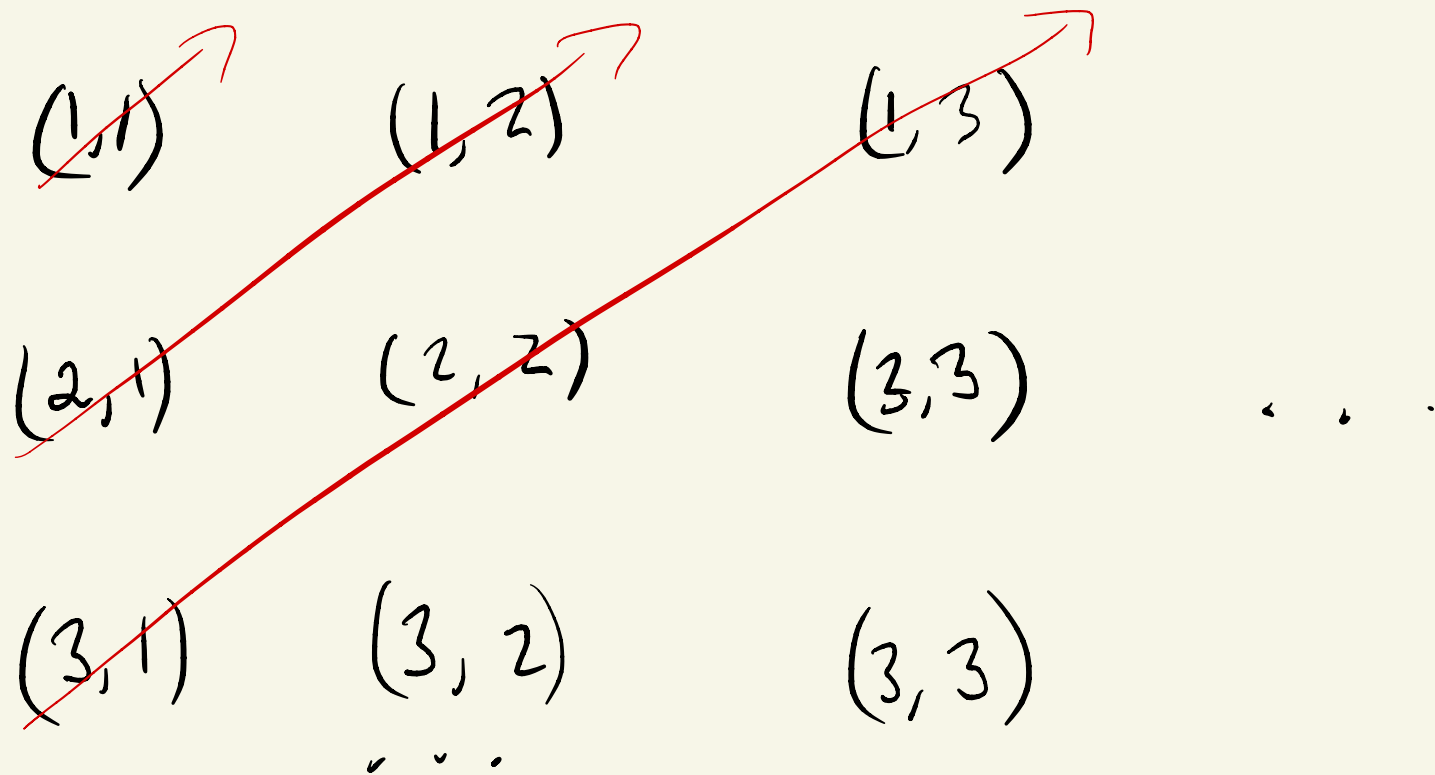
We write $x_n = \phi(n)$, and call the sequence $x_1, x_2, x_3, \dots$ or (x_n) .

*↑ slight difference
from Rudin's notation $\{x_n\}$.*

"Arrange A into a sequence" means choose a bijection $\mathbb{N} \rightarrow A$.

Theorem $\mathbb{N}^2$ is countable.

proof



We can arrange $\mathbb{N}^2$ into a sequence.

$(1,1), (2,1), (1,2), (3,1), (2,2), (1,3), \dots$

Theorem A subset of a countable set is at most countable.

Proof: Let $E \subset A$. Suppose E is infinite.

Arrange elements of A into a sequence $\{x_n\}$.

$x_1, x_2, x_3, x_4, x_5, \dots$

$n_1=1 \quad n_2=3 \quad n_3=4 \quad \dots$

Let $n_1 \in \mathbb{N}$ be the smallest natural number s.t. $x_{n_1} \in E$

Having chosen $n_1, \dots, n_k$, let n_{k+1} be the next smallest natural number s.t. $x_{n_{k+1}} \in E$. We have arranged E into a sequence (x_{n_k}) .

Conclusion: no cardinality between finite and countable.

Corrolary

- $A \hookrightarrow B$, B countable $\Rightarrow A$ at most countable
- $A \twoheadrightarrow B$, A countable $\Rightarrow B$ at most countable

Def (Indexed unions + intersections)

If I is a set, and E_α is a set for each $\alpha \in I$,

- $\bigcup_{\alpha \in I} E_\alpha = \{x \mid \exists \alpha \in I \text{ s.t. } x \in E_\alpha\}$
- $\bigcap_{\alpha \in I} E_\alpha = \{x \mid \forall \alpha \in I, x \in E_\alpha\}$

Theorem Let $\{E_n\}_{n \in \mathbb{N}}$ be a sequence of countable sets.

$\bigcup_{n \in \mathbb{N}} E_n$ is countable.

proof: Arrange each E_n into a sequence (x_{nk}) .

$\phi: \mathbb{N}^2 \longrightarrow \bigcup_{n \in \mathbb{N}} E_n$, $(n, k) \mapsto x_{nk}$ is surjective, and $\mathbb{Z}^2$ is countable,

so $\bigcup_{n \in \mathbb{N}} E_n$ is at most countable. $\bigcup_{n \in \mathbb{N}} E_n$ contains E_1 ,

so is not finite.

Theorem $\mathbb{Q}$ is countable.

Proof Let $E_n = \left\{ \frac{m}{n} \mid m \in \mathbb{Z} \right\}$

$$\mathbb{Q} = \bigcup_{n=1}^{\infty} E_n$$

$\mathbb{Q}$ is a countable union of countable sets.

Theorem $\mathbb{N}^l = \underbrace{\mathbb{N} \times \dots \times \mathbb{N}}_l$ is countable.

$\mathbb{N}$ countable by def. Assume $\mathbb{N}^{l-1}$ countable.

$\mathbb{N}^l \sim \bigcup_{n \in \mathbb{N}} \{n\} \times \mathbb{N}^{l-1}$ and $\{n\} \times \mathbb{N}^{l-1} \sim \mathbb{N}^{l-1}$ countable, so $\mathbb{N}^l$ countable.

Countable Sets

$\mathbb{N}$ $\mathbb{Q}$

$\mathbb{Z}$ $\mathbb{N} \times \dots \times \mathbb{N}$

$\mathbb{N} \times \mathbb{N}$ What about $\mathcal{P}(\mathbb{N})$?

Thm There is no surjection $A \rightarrow P(A)$ for any non-empty A .

Proof Suppose $\phi: A \rightarrow P(A)$ was a bijection.

Let $E \subset A$ be the subset $\{x \in A \mid x \notin \phi(x)\}$.

Suppose $E = \phi(x)$ for some x .

Suppose $x \in E$. Then, $x \notin \phi(x) = E$. Contradiction

Suppose $x \notin E$. Then $x \notin \phi(x)$ so $x \in E$. contradiction.

So x is neither in or not in E . Contradiction.

Theorem $\mathbb{R}$ is uncountable.

proof Suppose we had a bijection $\mathbb{N} \rightarrow \mathbb{R}$.

That is, a list $r_1, r_2, r_3, \dots$ of all real numbers.

Decimal expansions

$$r_1 = \textcircled{3}.1257\dots$$

$$r_2 = 100.\textcircled{0}010\dots$$

$$r_3 = 0.3\textcircled{3}33\dots$$

$$r_4 = 3.14\textcircled{0}5\dots$$

Let $r_i = n_{i0}.n_{i1}n_{i2}n_{i3}\dots$ be the decimal expansion of r_i .

Let $m_j = \begin{cases} 3 & \text{if } n_{jj} \neq 3 \\ 2 & \text{if } n_{jj} = 3 \end{cases}$ so that $m_j \neq n_{jj} \quad \forall j$.

$m_0, m_1, m_2, m_3, \dots$ is a real number not in our list.

Cardinalities so far

$$\emptyset < \underline{1} < \underline{2} < \dots < \mathbb{N} < \mathbb{R}$$

Q Where does $\mathcal{P}(\mathbb{N})$ fall in this order?

$$\mathcal{P}(\mathbb{N}) \sim \{\text{binary sequences } \mathbb{N} \rightarrow \{0, 1\}\}$$

Using binary expansion, can show $\mathcal{P}(\mathbb{N}) \sim \mathbb{R}$

Q Is there a set S with $\mathbb{N} < S < \mathbb{R}$?

This is Cantor's Continuum Hypothesis / First of Hilbert's 23 problems.

Undecidable.

Q are there bigger cardinalities?

Yes. $S < \mathcal{P}(S) < \mathcal{P}(\mathcal{P}(S)) < \dots$ for any

non-empty S .

Metric Spaces

Google maps biking times

	NYC	NewHaven	Springfield	providence
NYC	0	9	14	20
NewHaven	9	0	6	11
Springfield	14	6	0	7
Providence	20	11	7	0

This is a metric space with 4 points

Def A metric space is a set X with a function $d: X \times X \rightarrow \mathbb{R}_{\geq 0}$

s.t for all $p, q \in X$,

1) $d(p, q) = 0$ iff $p = q$ *non-degeneracy*

2) $d(p, q) = d(q, p)$ *symmetry*

3) $d(p, q) \leq d(p, r) + d(r, q)$ for all $r \in X$. *triangle inequality*

d is called a metric on X . Elements of X are called points.

Example: $\mathbb{R}$ with $d(x, y) = |x - y|$.
"standard metric"

Example: $\mathbb{R}$ with $d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$
"discrete metric"

Non-Example: $\mathbb{R}$ with $d(x, y) = (x - y)^2$

$$d(0, 1) = 1$$

$$d(0, \frac{1}{2}) + d(\frac{1}{2}, 1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$d(0, \frac{1}{4}) + \dots + d(\frac{3}{4}, 1) = \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} = \frac{1}{4}$$

Exotic Example:

2-adic metric on $\mathbb{Z}$:

$d(a, b) = 2^{-n}$ where 2^n the highest power of 2 dividing $a - b$.

$$d_{2\text{-adic}}(5, 6) = 1 \quad d_{2\text{-adic}}(4, 16) = \frac{1}{4}$$